

BIKENDRA THAPALIYA

Senior Software Engineer / Technical Lead / Solution Architect

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PROFESSIONAL SUMMARY

Solution-oriented engineering leader with **6+ years** architecting, leading, and delivering enterprise software in **C#/.NET and Angular** for banking, healthcare, telecom, and logistics clients. **2+ years** as Technical Lead / Solution Architect, leading teams of 3–6 through event-driven and microservices platforms — from requirement analysis and HLD/LLD/ADRs through code review, mentoring, and production delivery. Delivered real-time fraud monitoring, banking access-control, microservices warehouse-management, and telecom automation platforms on ASP.NET Core, Entity Framework Core, and containerised (Docker / Kubernetes) infrastructure with CI/CD via GitHub Actions and Jenkins. Currently designing a governed AI-assisted SDLC workspace to scale coding agents safely across engineering teams.

CORE COMPETENCIES

Leadership: Technical team leadership (teams of 3–6), mentoring, code review & best-practice enforcement, delivery ownership, risk escalation, stakeholder & client communication

Architecture: Solution architecture, microservices, event-driven architecture, RESTful API design, HLD/LLD, ADRs, technical roadmaps

Backend: C#, ASP.NET Core / .NET Core, .NET Framework, Entity Framework Core, ABP Framework, gRPC, SignalR, WorkflowCore, Hangfire

Frontend: Angular, React (portal delivery)

Data & Messaging: SQL Server, PostgreSQL, MongoDB, Redis, Apache Kafka, RabbitMQ, ksqldb, Apache Flink, Apache Druid, Elasticsearch/Kibana

DevOps & Delivery: Docker, Kubernetes/Helm, GitHub Actions, Jenkins, CI/CD pipelines, Agile/Scrum

Security & Integration: OAuth2/OpenID Connect, LDAP, third-party API & EDI integration

PROFESSIONAL EXPERIENCE

AMNIL Technologies Pvt. Ltd. — Kathmandu, Nepal

Jul 2022 – Present

Senior Software Engineer / Technical Lead / Solution Architect (Leadership scope since Dec 2023; promoted to Senior after ~3 months)

- Lead architecture and delivery across banking, telecom, healthcare, logistics, and IoT engagements; translate ambiguous business requirements into documented technical solutions (HLD/LLD, ADRs).
- Lead and mentor teams of 3–6 through project lifecycles; run design reviews and code reviews, own technical decisions, risk escalation, and stakeholder communication.
- Main technical lead for a ~23-month, multi-project client portfolio (Dec 2023 – Nov 2025) spanning warehouse, hospitality, and healthcare IoT tracking solutions — owning client communication, resourcing, and delivery while parallel-managing other engagements.
- **TransGuard** (real-time fraud monitoring, two banks): architected an event-driven platform on Kafka/ksqlDB/Flink/Druid processing **~20M transactions/month**, load-validated at 20,000 TPS; delivered dual detection (deterministic rules + behavioural profiling), pre-authorisation decisioning on the mobile channel, and end-to-end mule-account detection — 15 rules live in production across both banks under maker-checker governance.
- **ACMS** (banking access-control platform): architected a configurable access-control and workflow-orchestration system on ABP Framework with React admin/customer portals, OAuth2/OpenID Connect security, and WorkflowCore/Hangfire orchestration — a single reusable API layer onboarded by two banks, processing ~1,000 access cases/day.
- **Drop WMS v2** (microservices warehouse management): backend lead member designing service-owned database schemas, API specs, and UI enhancements across Administration, Inbound, Outbound, EDI, and Notification services (.NET Core, ABP, RabbitMQ, gRPC, Docker); migrated 50K+ records with zero data loss and built an EDI service reliably handling 1,000+ daily requests against a third-party logistics API.
- **Ncell Automated Recharge Scheduler:** led architecture and design of a WorkflowCore/Kafka-based scheduler achieving **99.9% delivery reliability** on 1,000–1,500 daily transactions; designed and delivered an OAuth2-secured REST job API and client integration guide.
- **IoT Asset Tracking:** designed RFID-based tracking (BSSID packet listening, RSSI base-station resolution) across hospital and hospitality clients, including ~4,000 tracked assets at one hospital site, reducing reported equipment loss by ~25% in the first quarter.
- Leading design of a **governed agentic SDLC workspace** (Claude Code + knowledge base + human-in-the-loop review gates) to run parallel AI-assisted development safely across engineering teams.
- Recognized with the company-wide “Extreme Ownership” award; sustained ≥3× peer-baseline delivery throughput through analysis-first requirement engineering.

Software Engineer

- Delivered features and bug fixes in a fast-paced team environment; gained direct exposure to client requirements and agile delivery under tight timelines.

Uranus Tech Nepal Pvt. Ltd. — Kathmandu, Nepal

Nov 2020 – Mar 2022

Software Engineer → Senior Software Engineer / Technical Lead

- Led modernisation of a desktop pension management system into a secure ASP.NET web application serving 125,000+ pensioners across 3 processing camps.
- Owned the full delivery loop end-to-end — requirements gathering, technical specification, coding, testing, client demos, and IIS deployment — while also acting as business analyst and product owner.
- Built microfinance modules (cheque, budget, loans, savings, KYC onboarding) and an HRIS web application on ASP.NET MVC / Clean Architecture; automated PowerShell-driven deployments, cutting release cycles from hours to under 15 minutes.
- Mentored junior developers from the second week of joining; progressed from intern to permanent within one week and to senior/technical lead within a few months.

SELECTED PROJECTS

TransGuard — Real-Time Transaction Fraud Monitoring

MBL Bank & NMB Bank

- Kafka / ksqldb / Flink / Druid streaming platform with a signal-to-case pipeline, severity scoring, SLA tracking, and auto-routing; Angular analyst portal on ASP.NET Core, deployed on Kubernetes via Helm.

ACMS — Access Control Management System

MBL Bank & NMB Bank

- ABP Framework platform with ~40 RBAC permission groups, WorkflowCore staged approvals (40+ reusable steps), Hangfire automation, and LDAP authentication; React Admin and Customer/Operations portals.

Drop WMS v2 — Microservices Warehouse Management

St Luke's / Schenker / Applied Materials

- Multi-tenant microservices rollout across three logistics sites, part of a broader migration of 50K+ records with zero data loss.

Ncell Automated Recharge Scheduler

Ncell Telecom

- Idempotent, Redis-backed event processing with Polly retry policies and Kafka reconnection logic; OAuth2-secured job API exposing scheduled, Kafka-triggered, and auto-pack-purchase recharge jobs.

Governed Agentic SDLC Workspace

Internal Initiative

- Design → Plan → Develop → Review → Deploy → Monitor pipeline built around Claude Code, spec-driven development, and human-in-the-loop review gates enabling parallel autonomous task execution.

EDUCATION

Bachelor of Engineering (BE), Information Science and Engineering

2015 – 2019

R R Institute of Technology, Bengaluru, India

Data Science & Machine Learning Training (self-directed)

2020 – 2021

DataCamp — Python, Data Analysis, Machine Learning Fundamentals, Model Evaluation

CERTIFICATIONS

DataCamp — Data Analyst, Data Scientist, Machine Learning Scientist, Python Programmer

COMMUNITY & PUBLICATIONS

- “Outcome Ownership Era: How AI Is Turning Software Inside Out” — LinkedIn Pulse
- “Your Codebase Is Not Your Most Valuable Asset — Your Knowledge Base Is” — Medium
- “Agentic Development Workflow: From Raw Requirements to Production Code Without Losing Control” — Medium